
CprE 458/558: Real-Time Systems

RMS, EDF Schedulers (contd) --
Exact Analysis

Exact Analysis (necessary & sufficient)

Critical Zone Theorem:

For a set of independent periodic tasks, if a task T_i meets its first deadline $d_i \leq p_i$, when all other higher priority tasks are started (ie., ready) at the same time, then it meets all its future deadlines with any other task start times.

Completion Time Test

Let there be n tasks ordered in decreasing priority. Consider any task T_i . The workload over $[0, t]$ (for arbitrary $t > 0$) due to all tasks of equal or higher priority than T_i is given by

$$W_i(t) = \sum_{j=1}^i c_j \left\lceil \frac{t}{p_j} \right\rceil$$

The term $\left\lceil \frac{t}{p_j} \right\rceil$ represents the number of times task T_j arrives in time t , and therefore $c_j \left\lceil \frac{t}{p_j} \right\rceil$ represents its computational demand in time t .

Completion Time Test (Contd.)

Suppose that task T_i completes its execution exactly at time t . This means that the total cumulative demand from the i tasks up to time t , $W_i(t)$, is exactly equal to t , that is, $W_i(t) = t$. A method for finding the completion time of task T_i , that is, the time at which $W_i(t) = t$, is known as completion time test.

Completion Time Test (Contd.)

$$\begin{aligned} \text{Set } t_0 &= \sum_{j=1}^i c_j \\ t_1 &= W_i(t_0) \\ t_2 &= W_i(t_1) \\ t_3 &= W_i(t_2) \\ &\vdots \\ t_k &= W_i(t_{k-1}) \\ \text{Stop when } W_i(t_k) &= t_k \end{aligned}$$

A task T_i is schedulable if $W_i \leq d_i$, where $W_i(t)=t$. An entire task set is schedulable if this condition holds for all the tasks in the set.

Completion Time Test — Example

Task T_1 : $c_1 = 20$; $p_1 = 100$; $d_1 = 100$.

Task T_2 : $c_2 = 30$; $p_2 = 145$; $d_2 = 145$.

Task T_3 : $c_3 = 68$; $p_3 = 150$; $d_2 = 150$.

This task set fails the (utilization-based) schedulability test for RMS. So, perform completion time test for T_1 , T_2 , T_3 . Task T_3 's completion time test is as follows.

$$t_0 = c_1 + c_2 + c_3 = 20 + 68 + 30 = 118.$$

$$t_1 = W_3(t_0) = 2c_1 + c_2 + c_3 = 40 + 68 + 30 = 138.$$

$$W_3(t_1) = 2c_1 + c_2 + c_3 = 40 + 68 + 30 = 138 = t_1.$$

Task T_3 is schedulable!; Tasks T_1 and T_2 too.

Deadline monotonic scheduling (DMS)

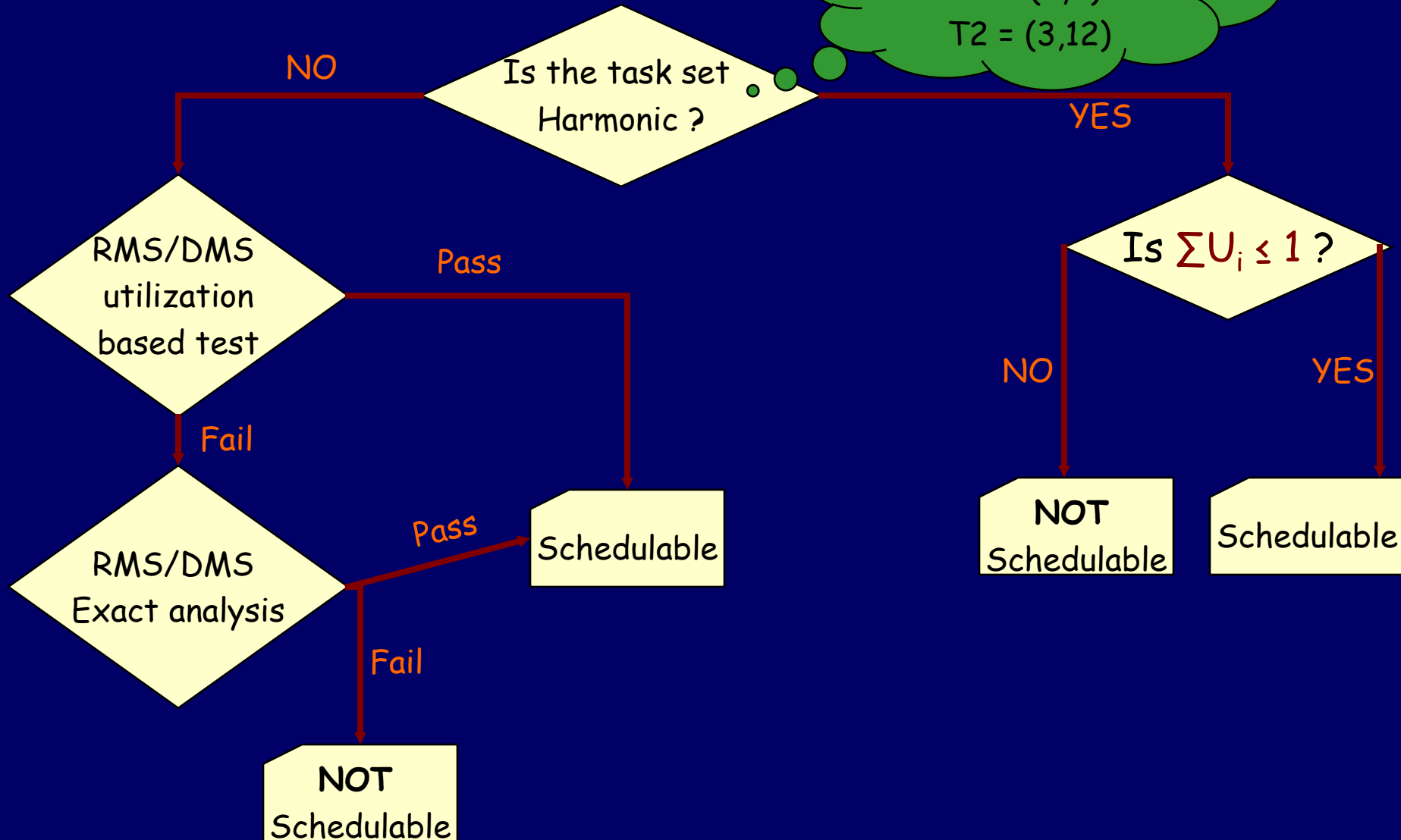
- Task T_i : (c_i, p_i, d_i) ; with relative deadline $d_i \leq p_i$
[Please note: “ d_i ” is a fixed parameter, whereas the “absolute deadline” used in EDF is a dynamic parameter]
- Assigns priority based on d_i ; smaller the d_i , higher the priority
- Similar to RMS utilization-based schedulability test, except c_i/d_i used instead of c_i/p_i
$$\sum C_i/d_i \leq n(2^{1/n} - 1)$$
- Similar to RMS exact analysis, except the ordering of tasks is based on d_i instead of p_i
- Example: (c_i, p_i, d_i) : $(3, 20, 7)$, $(2, 5, 4)$, $(2, 10, 9)$. This task set is schedulable even though $\text{Sum}(c_i/d_i) > 1$.
- DMS is also an optimal fixed-priority scheduling algorithm; it is a generalization of RMS.

RMS/DMS Schedulability

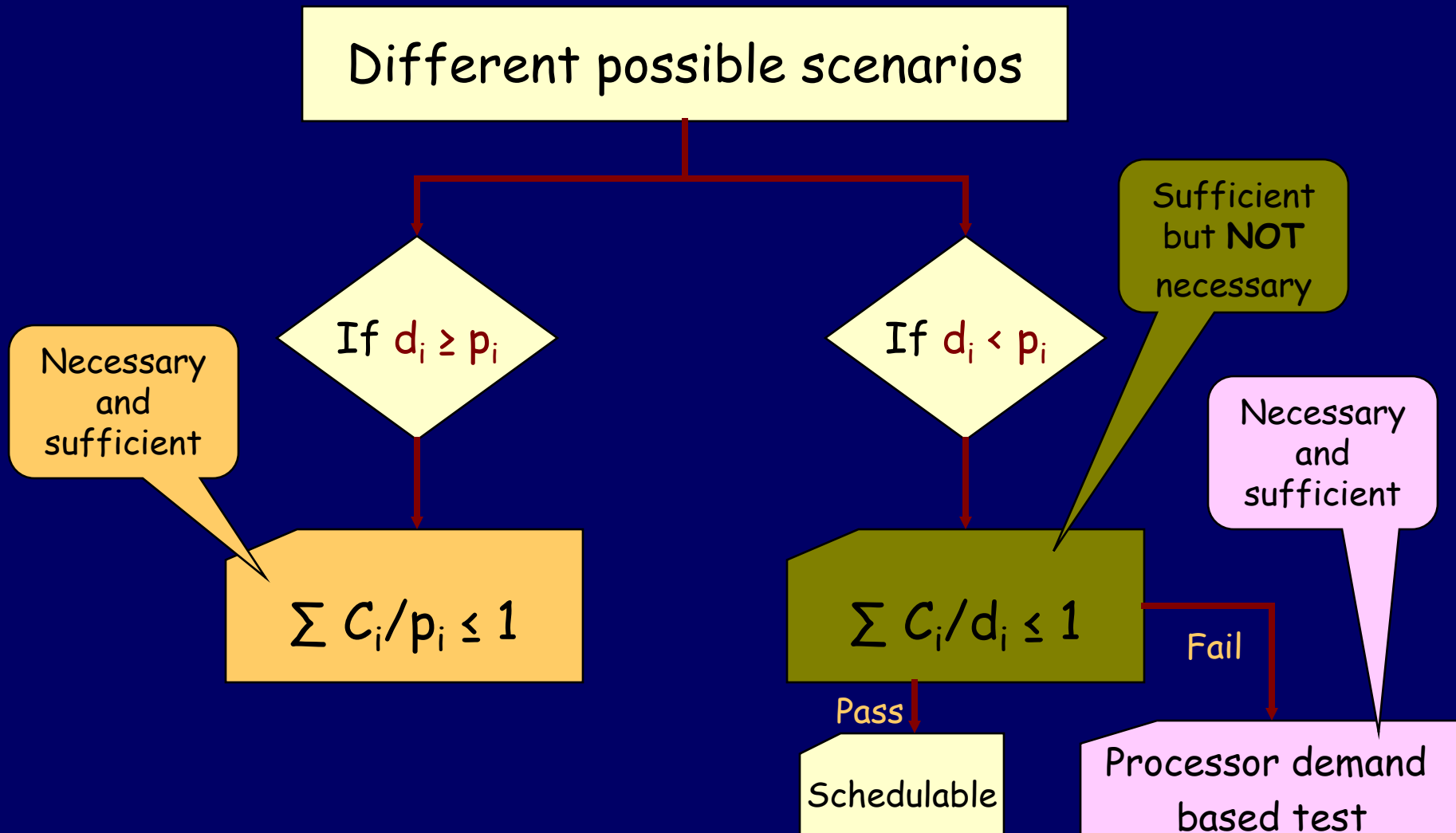
Example task set:

$T1 = (2,4)$

$T2 = (3,12)$



EDF Revisited: Schedulability test



Periodic task scheduling - summary

$$D_i = P_i$$

$$D_i \leq P_i$$

Static
priority

RMS

Processor
Utilization test

DMS

Exact Analysis

Dynamic
priority

EDF/LLF

Processor
utilization test
($U \leq 1$)

EDF

Processor
demand based
test (**not covered**)